International and Monetary Economics

Course Code: 30054

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Lecture 14

Roadmap

1. How to Fix FX?
2. Stabilization policies
3. Central bank interventions
4. Sterilized interventions

How to Fix FX?

How to fix FX?

- How?
  - A central bank stands ready to buy or sell any amount of foreign currencies at the announced fixed rate (assuming free capital mobility, of course)

- What does it imply?
  - If $E = E^e$, then

$$R = R^* + \underbrace{\frac{E^e - E}{E}}_{=0}$$

How to fix FX?

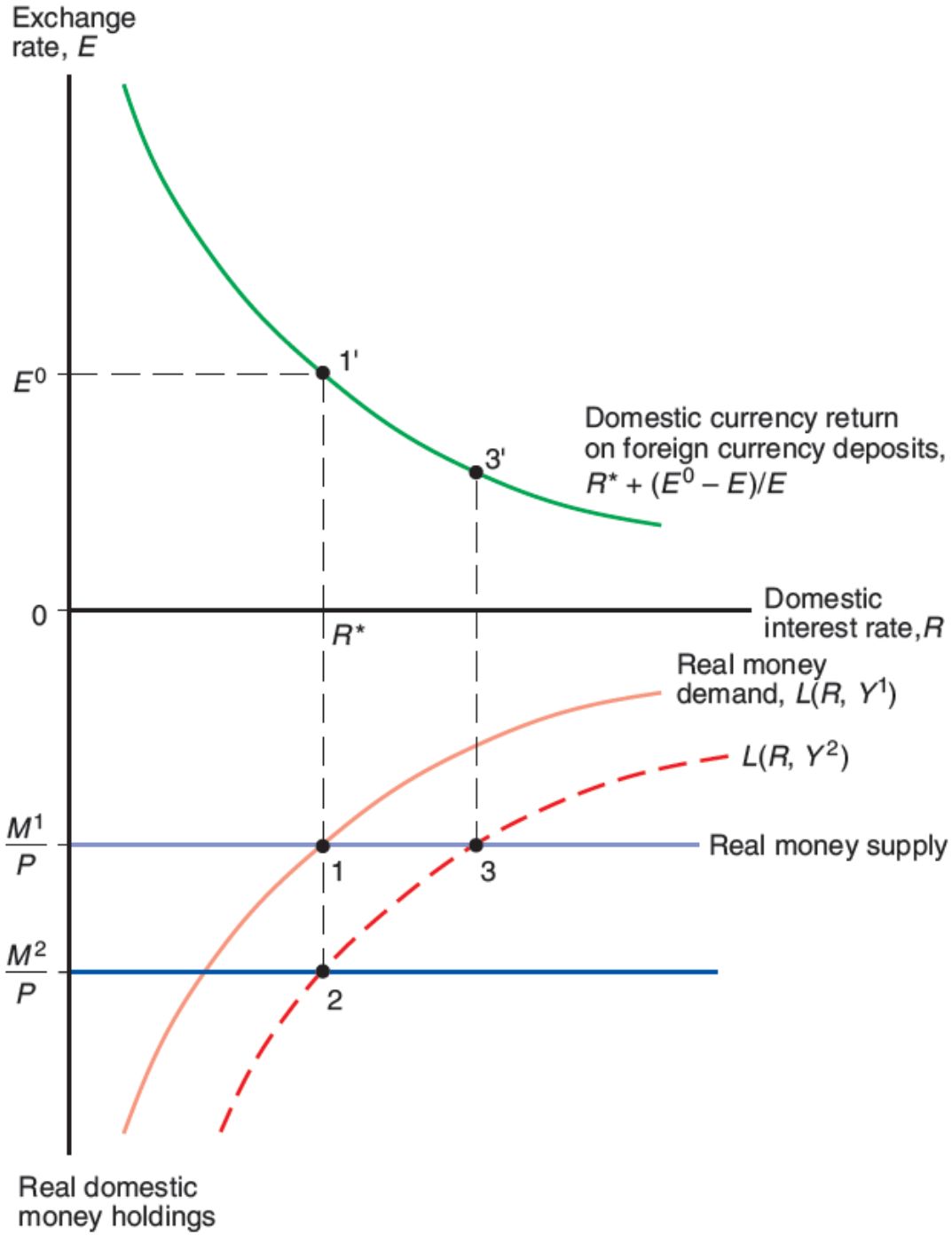
Money Market

- Money market

$$\frac{M^s}{\bar{P}} = L(R, \bar{Y})$$

  – $\bar{P}$ is fixed in the short run, $\bar{Y}$ is fixed for simplicity

- To fix FX, the CB needs to fix *R* by choosing M^s
  – Any change that affects money market must be accompanied by a change in money supply M^s



How to fix FX?

Example

- An increase in domestic output must be met by an increase in money supply through purchases of foreign **and/or** domestic assets (*b/c they are perfect substitutes in this model*)

Stabilization Policy

Stabilization Policy

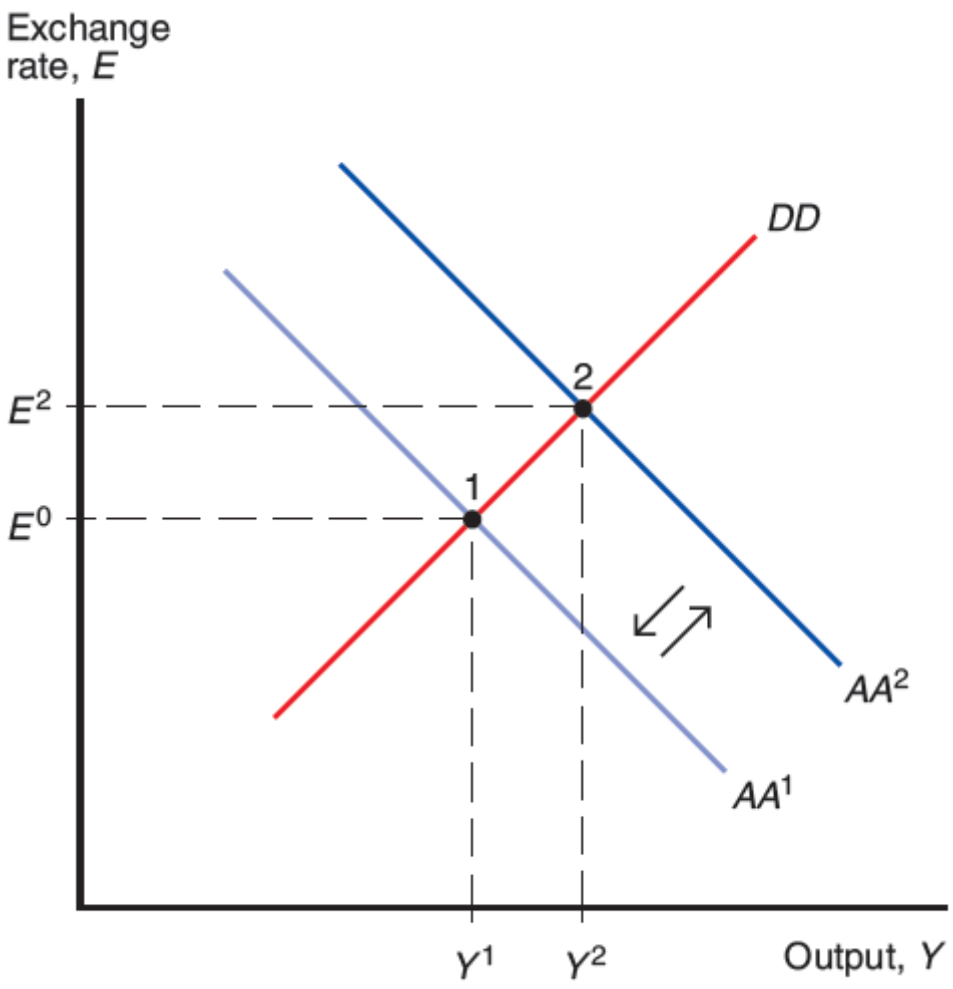
1. ~~Monetary policy~~

2. **Fiscal policy**

3. *Exchange rate policy*

Stabilization Policy

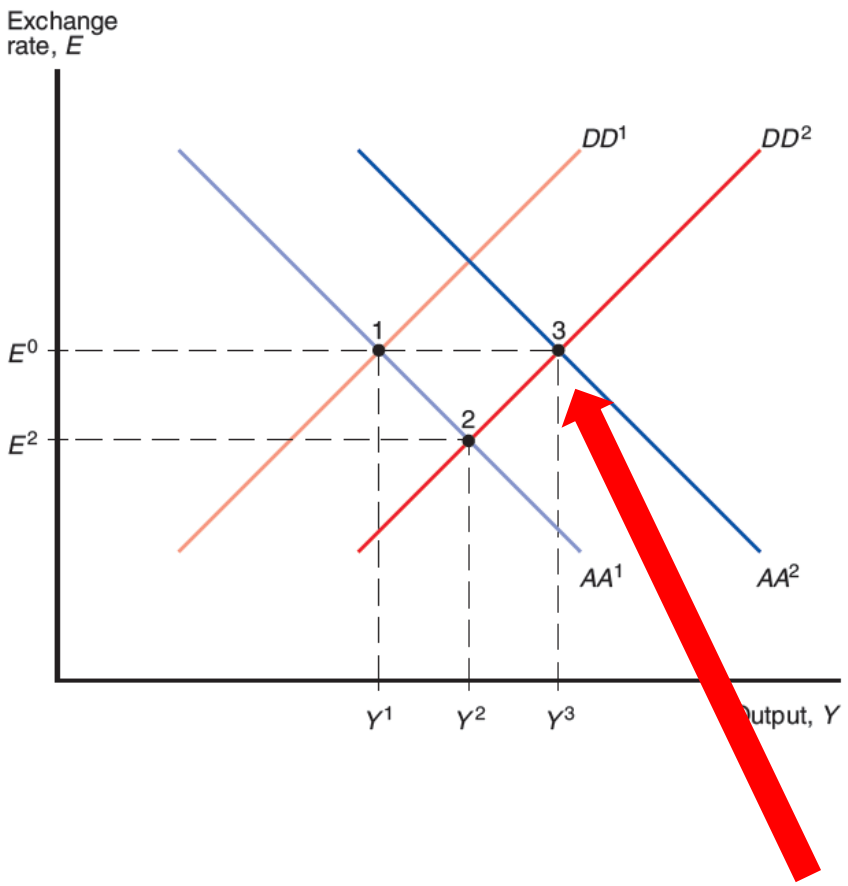
Monetary Policy



Any attempt at stimulating the economy by an increase in money supply must be met by a decline in money supply to sustain fixed FX

Stabilization Policy

Fiscal Policy



- Fiscal expansion shifts DD to the right
- FX appreciates to E^2
- To prevent this appreciation, the CB increases money supply (e.g., by purchasing foreign currency)

Fiscal policy has a stronger effect on output under fixed FX

Stabilization Policy

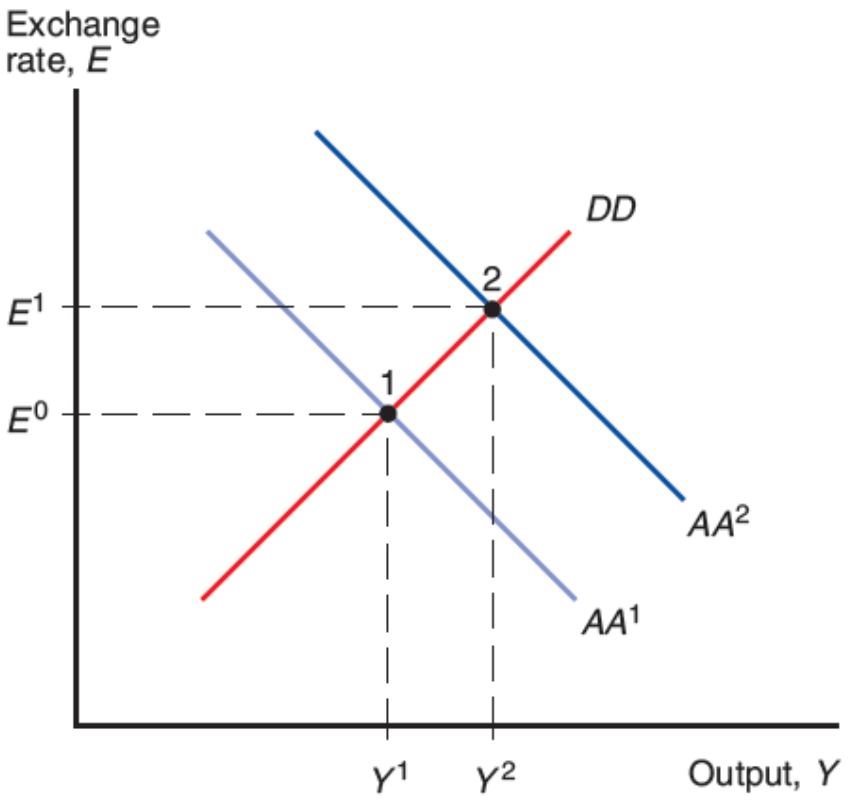
Exchange Rate Policies

Language

- $E \uparrow$
  - Floating FX
    - Verb: currency depreciated
    - Noun: depreciation
  - Fixed FX:
    - Verb: currency was devalued
    - Noun: devaluation
- $E \downarrow$
  - Floating FX
    - Verb: currency appreciated
    - Noun: appreciation
  - Fixed FX:
    - Verb: currency was revalued
    - Noun: revaluation

Stabilization Policy

Exchange Rate Policy



- A rise in E from E^0 to E^1
  - movement along DD
- Higher Y increases R, money supply needs to increase to reduce R to its R^* level
  - AA shifts to the right

Stabilization Policy

Exchange Rate Policy

- Devaluation leads to
  - A rise in output
  - An expansion in the money supply
  - A rise in official reserves (if gov. purchases foreign assets with new money)
- It is chosen by governments to
  - Fight domestic unemployment
  - Improve the current account
  - Affect the central bank's foreign reserves

Central Bank Interventions

Central Bank Interventions

Balance Sheet

Assets		Liabilities	
Foreign assets	$1,000	Deposits held by private banks	$500
Domestic assets	$1,500	Currency in circulation	$2,000

- Total assets = total liabilities + net worth
  - Note that the above table assumes net worth equals zero
- If net worth is constant, then
  - any central bank purchase of assets automatically results in an increase in the domestic money supply
  - any central bank sale of assets automatically causes the money supply to decline

Central Bank Interventions 1

A Change in the **Size** of the Balance Sheet

Central Bank Balance Sheet

Assets		Liabilities	
Foreign assets	$1,000	Deposits held by private banks	$500
Domestic assets	$1,500	Currency in circulation	$2,000

Central Bank Balance Sheet after $100 Foreign Asset Sale (Buyer Pays with Currency)

Assets		Liabilities	
Foreign assets	$900	Deposits held by private banks	$500
Domestic assets	$1,500	Currency in circulation	$1,900

Central Bank Interventions 2

A Change in the **Asset Composition** of the Balance Sheet

- Central banks sometimes carry out equal foreign and domestic asset transactions in opposite directions to nullify the impact of their foreign exchange operations on the domestic money supply (sterilized intervention)

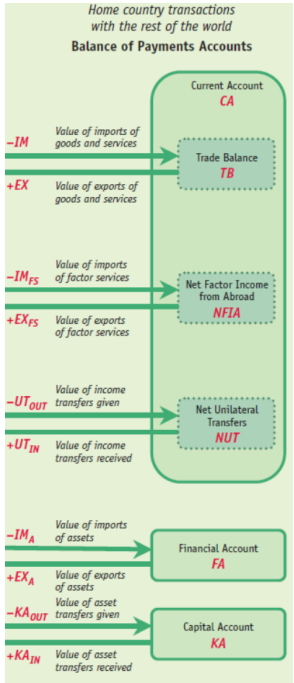
Central Bank Balance Sheet before Sterilized $100 Foreign Asset Sale

Assets		Liabilities	
Foreign assets	$1,000	Deposits held by private banks	$500
Domestic assets	$1,500	Currency in circulation	$2,000

Assets		Liabilities	
Foreign assets	$900	Deposits held by private banks	$500
Domestic assets	$1,600	Currency in circulation	$2,000

Central Bank Interventions

The Balance of Payments



- Recall

$$FA + CA + KA = 0$$

- Expand

$$FA(no\ CB) + FA(CB) + CA + KA = 0$$

  – FA(no CB) is financial account balance excluding the central bank
  – FA(CB) is a financial account balance coming from the central bank

- Rearrange

$$FA(no\ CB) + CA + KA = -FA(CB) = \Delta OR$$

  – ΔOR is a change in **official reserves**

Sterilized Interventions

Sterilized Interventions

- Sterilized interventions are neutral in our discussion so far
  - b/c domestic and foreign bonds are perfect substitutes when their interest rates are equal
  - It is the money supply that matters!

- Prominent channels of sterilized interventions non-neutrality (the same two channels are relevant for quantitative easing)
  1. Portfolio balance channel
  2. Signaling channel

Portfolio Balance Channel

- Domestic and foreign bonds are imperfect substitutes in reality
  - difference in risk
  - market segmentation: not everyone can trade foreign bonds

- By changing the supply of government bonds available to private agents, the central bank can influence the interest rate and, hence, the exchange rate

Portfolio Balance Channel

- Perfect substitutability

$$R = R^* + \frac{E^e - E}{E}$$

- Imperfect substitutability

$$R = R^* + \frac{E^e - E}{E} + \rho$$

  – where ρ is a wedge that represents differences in the amount of risk in domestic and foreign bonds

  – Where does it come from?

Portfolio Balance Channel

- Total demand for domestic bonds

$$B^d = B^d\left(R - \frac{E^e - E}{E} - R^*\right)$$

- Bonds market equilibrium

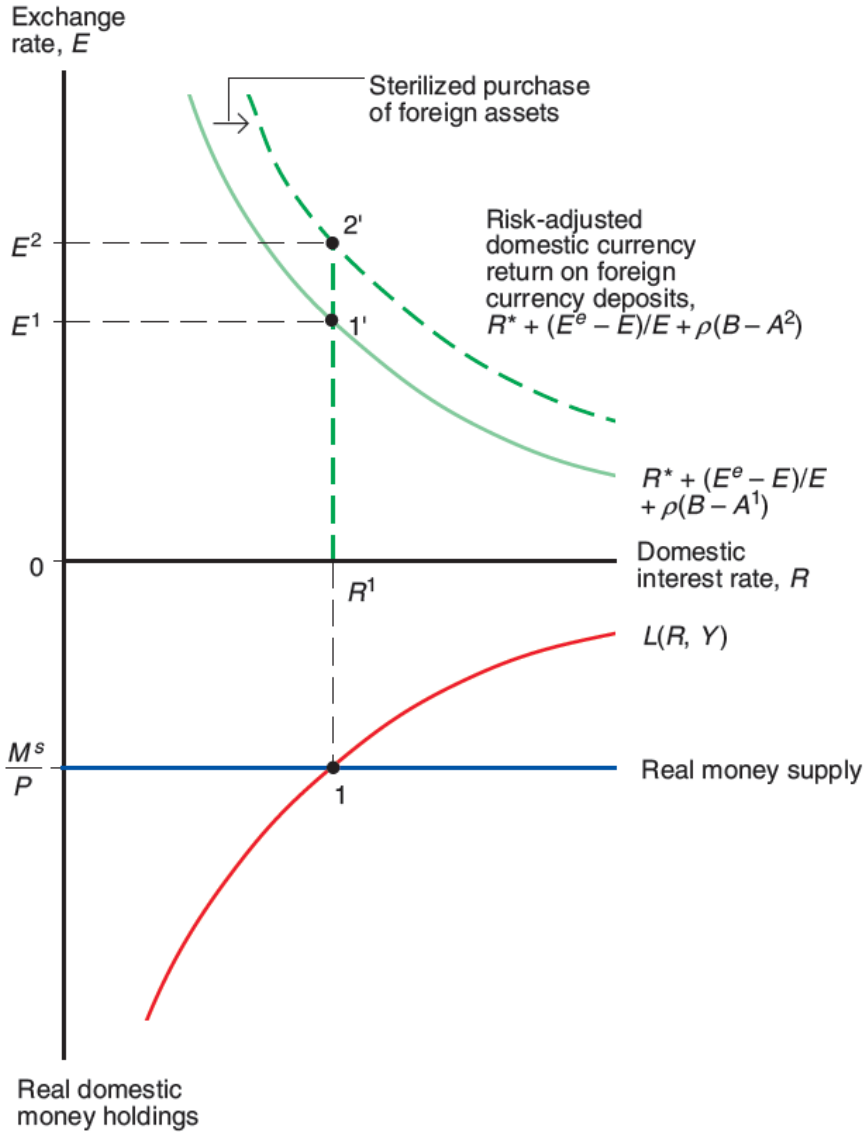
$$\underbrace{B - A}_{\substack{\text{net supply of} \\ \text{government bonds}}} = B^d\left(R - R^* - \frac{E^e - E}{E}\right)$$

$$\Rightarrow R - R^* - \frac{E^e - E}{E} = \rho(B - A)$$

  – B is the total supply of government bonds

  – A is the amount of government bonds held by the central bank

Sterilized Interventions



- A purchase of foreign bonds accompanied by a sale of domestic government bonds increases the risk premium
- E depreciates (if the expected exchange rate is unchanged)

Signaling Channel

- Sterilized purchases of foreign assets are interpreted as signals of the central bank's intent to depreciate the exchange rate in the future via a standard money supply increase. Future expected depreciation causes depreciation today b/c the exchange rate is like an asset price.

- These purchases make the commitment to depreciate FX in the future more credible
  - the CB will get a profit on its portfolio of foreign assets if the exchange rate depreciates

Are sterilized interventions effective?

- Big disagreement among economists!
- Difficult to interpret existing estimates
  - Reverse causality that introduces a negative bias: central banks purchase foreign bonds when the FX appreciates
  - Bond markets in advanced economies are very large, making it hard to influence the interest rate through purchases (this is not the case for developing countries)
- Recent evidence indicates that sterilized interventions are effective
  - Domiguez-Fatum-Vacek (2013),Chamon-Garcia-Souza (2017)